

Theorem (Jensen's Inequality). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex, i.e.*

$$\forall x, y \in \mathbb{R}, \forall t \in [0, 1] : \quad f(tx + (1-t)y) \leq tf(x) + (1-t)f(y). \quad (\text{C})$$

Let $x_1, \dots, x_n \in \mathbb{R}$ and let weights $\lambda_1, \dots, \lambda_n$ satisfy

$$\lambda_i \geq 0 \ (i = 1, \dots, n), \quad \sum_{i=1}^n \lambda_i = 1.$$

Then

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (\text{J})$$

Proof. **1. Degenerate cases.**

If $\lambda_k = 1$ and $\lambda_i = 0$ for $i \neq k$, then $\sum_{i=1}^n \lambda_i x_i = x_k$ and $\sum_{i=1}^n \lambda_i f(x_i) = f(x_k)$; (J) reduces to the identity $f(x_k) \leq f(x_k)$.

If some weight equals 0, the corresponding index can be omitted from both sides of (J). Hence we may assume that at least two weights are positive.

2. Binary case ($n = 2$).

Let $\lambda_1, \lambda_2 \geq 0$ with $\lambda_1 + \lambda_2 = 1$. Applying (C) with $t = \lambda_1$, $x = x_1$, $y = x_2$ yields

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2),$$

which is (J) for $n = 2$.

3. Inductive step.

Assume (J) holds for any collection of $n - 1$ points ($n \geq 3$). Let $x_1, \dots, x_n$ and non-negative weights $\lambda_1, \dots, \lambda_n$ satisfy $\sum_{i=1}^n \lambda_i = 1$.

Set

$$L := \sum_{i=1}^{n-1} \lambda_i, \quad 1 - L = \lambda_n.$$

Since the λ_i are non-negative, $0 \leq L \leq 1$.

If $L = 0$ (resp. $L = 1$), we are back to the trivial single-weight case or to the induction hypothesis applied to the first $n - 1$ points. Thus we may assume $0 < L < 1$.

Define normalized weights for the first $n - 1$ points:

$$\mu_i := \frac{\lambda_i}{L} \quad (i = 1, \dots, n-1),$$

so that $\mu_i \geq 0$ and $\sum_{i=1}^{n-1} \mu_i = 1$. Let

$$y := \sum_{i=1}^{n-1} \mu_i x_i.$$

Induction hypothesis applied to $\{x_i\}_{i=1}^{n-1}$ gives

$$f(y) \leq \sum_{i=1}^{n-1} \mu_i f(x_i). \quad (\text{IH})$$

The full convex combination can be rewritten as

$$\sum_{i=1}^n \lambda_i x_i = L y + (1 - L) x_n. \quad (*)$$

Applying convexity (C) to the pair (y, x_n) with parameter $t = L$ yields

$$f(Ly + (1 - L)x_n) \leq L f(y) + (1 - L)f(x_n). \quad (C2)$$

Using (IH) and the definition of μ_i ,

$$L f(y) = L \sum_{i=1}^{n-1} \mu_i f(x_i) = \sum_{i=1}^{n-1} \lambda_i f(x_i). \quad (*1)$$

Now

$$\begin{aligned} f\left(\sum_{i=1}^n \lambda_i x_i\right) &\stackrel{(*)}{=} f(Ly + (1 - L)x_n) \\ &\leq L f(y) + (1 - L)f(x_n) \\ &\stackrel{(*1)}{=} \sum_{i=1}^{n-1} \lambda_i f(x_i) + \lambda_n f(x_n) \\ &= \sum_{i=1}^n \lambda_i f(x_i), \end{aligned}$$

which is precisely (J) for n points. Thus, by induction, (J) holds for every integer $n \geq 1$.

4. Equality condition.

In the binary inequality (C) equality holds iff either $t = 0$ or $t = 1$ (one weight is zero) or the restriction of f to the segment $[x, y]$ is affine. In the inductive proof we used (C) once, with $t = L$ and points y and x_n . Hence equality in (J) occurs precisely when

1. $L = 0$ or $L = 1$ (all weight concentrated on a single point), or
2. f is affine on the segment $[y, x_n]$ and equality already held in the induction hypothesis for the first $n - 1$ points.

Iterating this observation shows that equality holds iff either all but one weight are zero, or f is affine on the convex hull of the points with positive weight. In particular, if all such points coincide or if f is globally affine ($f(t) = at + b$), equality holds for every choice of $\{x_i\}, \{\lambda_i\}$.

□